OPERATING SYSTEMS

UNIT II – PROCESS MANAGEMENT & CPU SCHEDULING

SECTION A – 2-MARK QUESTIONS (3-MARK ANSWER FORMAT)

Answers, examples and simple diagrams

Q1. What are the different states of a process?

Definition:

A Process State represents the current condition of a process while it is running in a computer.

Explanation:

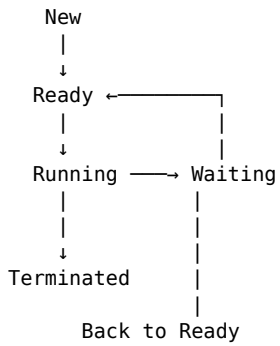
- New: Process is being created.
- Ready: Process is waiting for CPU.
- Running: Process is executing in CPU.
- Waiting: Process is waiting for an event or I/O operation.
- Terminated: Process has completed its work.

Note:

In the exam, draw the standard process-state diagram with Ready → Running → Waiting → Ready transitions.

Example: When we open a game, it is created, waits for CPU, runs and finally closes.

Diagram:



Q2. Differentiate Preemptive and Non-Preemptive Scheduling.

Definition:

CPU Scheduling is the method of selecting a process to execute in the CPU.

Differences:

Preemptive Scheduling: CPU can be taken from a process; process can be stopped in between; Example: Round Robin. Non-Preemptive Scheduling: CPU cannot be taken forcefully; process continues until it finishes or waits; Example: FCFS.

Example: Preemptive: Teacher stops one student and gives a chance to another. Non-Preemptive: One student completes the answer before another gets a chance.

Diagram:

```
Preemptive:
Process A → Stop → Process B

Non-Preemptive:
Process A → Complete → Process B
```

Q3. What is a Process Control Block (PCB)? What does PCB contain?

Definition:

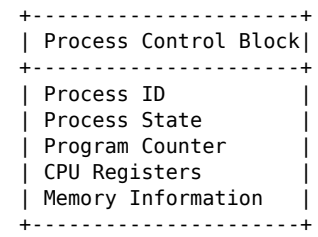
PCB is a data structure maintained by the Operating System. It stores all important information about a process.

Explanation:

- Process ID: Unique number of a process.
- Process State: Current state of the process.
- Program Counter: Address of the next instruction.
- CPU Registers: Stores CPU information.
- Memory Information: Details about memory used by the process.

Example: PCB is like a student record containing name, roll number and marks.

Diagram:



Q4. List any two CPU Scheduling Algorithms.

Definition:

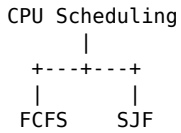
CPU Scheduling Algorithms are methods used by the OS to select which process gets the CPU.

Explanation:

- FCFS (First Come First Serve): The process that comes first gets the CPU first.
- SJF (Shortest Job First): The process with the shortest execution time gets the CPU first.

Example: In a ticket counter, the person who comes first gets served first (FCFS).

Diagram:



Q5. Differentiate between Process and Thread.

Definition:

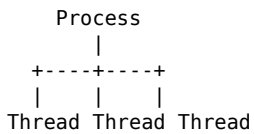
A Process is a program in execution, whereas a Thread is a small unit of execution inside a process.

Differences:

Process: A program in execution; has its own memory space; creation takes more time. Thread: Small unit of a process; shares memory with other threads of the same process; creation takes less time.

Example: A browser is a process. Downloading a file and playing a video can work as separate threads.

Diagram:



```
      1     2     3
```

Q6. What is the role of a Dispatcher in CPU Scheduling?

Definition:

A Dispatcher is an OS component that gives control of the CPU to the process selected by the CPU Scheduler.

Explanation:

- It performs context switching.
- It switches the CPU to user mode.
- It starts the selected process.

Example: A teacher selects a student, and the dispatcher gives that student a chance to speak.

Diagram:

```
    CPU Scheduler
          |
          ↓
     Dispatcher
          |
          ↓
    Selected Process
          |
          ↓
          CPU
```

Q7. Define Process.

Definition:

A Process is a program that is currently running in a computer.

Explanation:

- A process contains program instructions.
- It uses CPU and memory.
- Each process has a unique Process ID (PID).

Example: When we open Calculator, it becomes a process.

Diagram:

```
  Program
     |
     | Execute
     ↓
   Process
     |
     ↓
     CPU
```

Q8. Define Schedulers.

Definition:

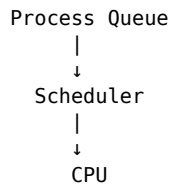
Schedulers are OS programs that select processes from different queues and decide which process should be executed.

Explanation:

- They select processes for execution.
- They manage process queues.
- They help in efficient CPU usage.

Example: A teacher selects students one by one to answer questions.

Diagram:



Q9. What are the types of Schedulers?

Definition:

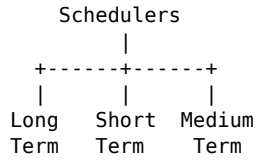
Schedulers are classified into three types based on their work.

Explanation:

- Long-Term Scheduler: Selects jobs and brings them into memory.
- Short-Term Scheduler: Selects a ready process and gives it to the CPU.
- Medium-Term Scheduler: Temporarily removes processes from memory and brings them back later.

Example: These schedulers manage which jobs enter, run and temporarily leave the system.

Diagram:



Q10. Define Thread.

Definition:

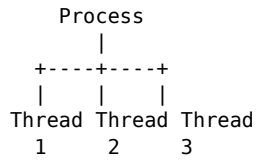
A Thread is the smallest unit of execution inside a process. It is also called a lightweight process.

Explanation:

- A process can have multiple threads.
- Threads share the memory of their process.
- They help perform multiple tasks at the same time.

Example: In a browser, one thread can play a video while another downloads a file.

Diagram:



Q11. What is a Multiprocessor System?

Definition:

A Multiprocessor System is a computer system that contains two or more CPUs (processors) working together.

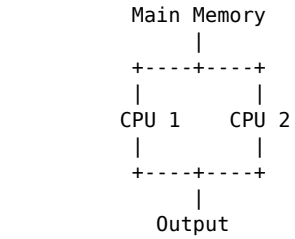
Explanation:

- It uses multiple processors.
- It can execute multiple tasks faster.

- It improves system performance.

Example: A computer with a multi-core processor.

Diagram:



Q12. Define Context Switching.

Definition:

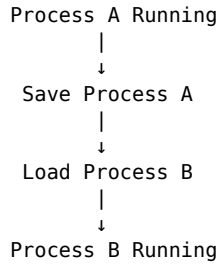
Context Switching is the process of saving the current process information and loading another process information into the CPU.

Explanation:

- The OS saves the current process state in its PCB.
- It selects another process.
- It loads the new process information into the CPU.

Example: While listening to music, we open a browser. The CPU switches between tasks.

Diagram:



Q13. List the Thread Libraries.

Definition:

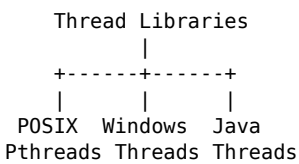
Thread Libraries are collections of functions used to create and manage threads in a program.

Explanation:

- POSIX Pthreads: Used in UNIX/Linux systems.
- Windows Threads: Used in Windows operating systems.
- Java Threads: Used in Java programming.

Example: Java programs use Java Thread Library to create multiple threads.

Diagram:



Unit II – Quick Revision

Q.No	Question	Easy Keywords
1	Process States	New, Ready, Running, Waiting, Terminated
2	Scheduling	Preemptive, Non-Preemptive
3	PCB	PID, State, Counter, Registers
4	CPU Algorithms	FCFS, SJF
5	Process vs Thread	Independent, Lightweight
6	Dispatcher	Context Switch, CPU Control
7	Process	Program in Execution
8	Schedulers	Select Processes
9	Types	Long, Short, Medium
10	Thread	Smallest Unit
11	Multiprocessor	Two or More CPUs
12	Context Switching	Save and Load
13	Thread Libraries	POSIX, Windows, Java